Python Engineering - Class 09 Theory Guide

Modules and Imports

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Why modules

As scripts grow, keeping everything in one file increases coupling and makes reuse/testing harder.

Relationship / equation

```
program = modules with responsibilities
```

Connection to code

EX17 separates physics, processing and orchestration.

2. Import execution model

When Python imports a module, top-level statements execute once to create the module namespace.

Relationship / equation

```
import -> execute definitions/top-level -> module object
```

Connection to code

This is why calculation demos should usually be placed under a main guard.

3. __name__ mental model

The same file can act as an executable script or as an importable module.

Relationship / equation

```
__name__ == '__main__' only for direct execution
```

Connection to code

The guard selects whether application-level execution should start.

4. Data flow between modules

Modules communicate through function inputs and returned outputs, not through hidden printing or global state.

Relationship / equation

```
main -> function(data) -> returned data
```

Connection to code

Returning results makes modules reusable by tests and future applications.

5. Architecture before scale

The goal is not to create many files for their own sake. Split only when responsibilities become meaningful.

Relationship / equation

```
cohesion high; coupling low
```

Connection to code

EX17 establishes the architecture later reused for engineering toolkits and larger projects.

Manual validation method

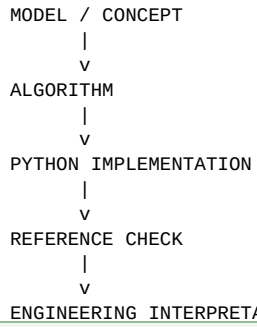
For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Modules establish the architecture used by every portfolio project.

Final summary



> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.